

Problem Set 4

Rahimeh Durrani

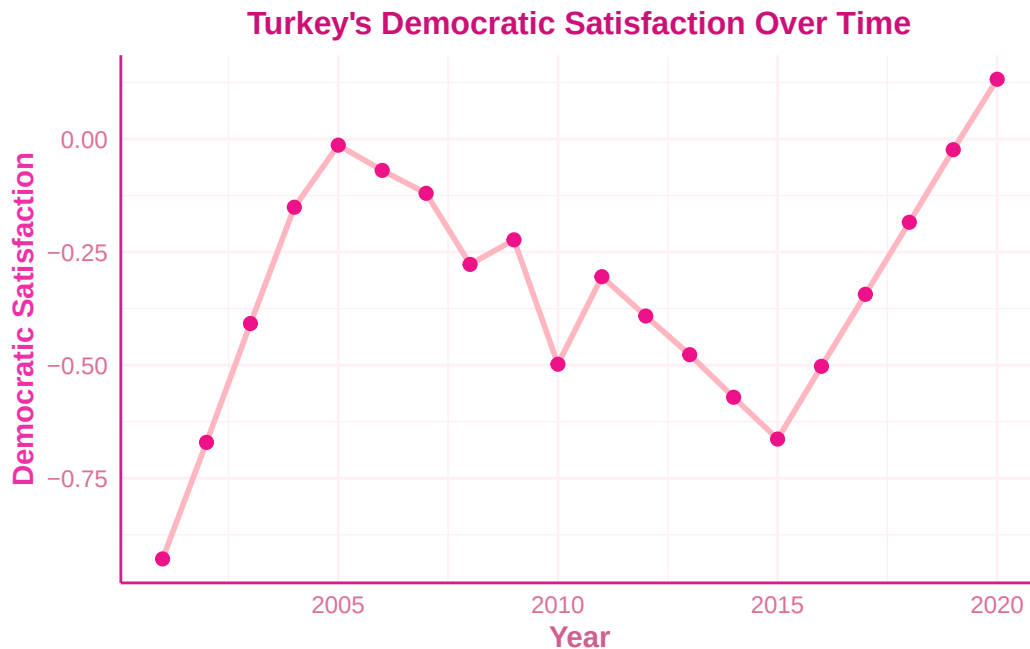
2026-01-11

Question 1: What are the dimensions of the dataset? Plot Turkey's democratic satisfaction score over time. What do you observe? Show your code.

```
read.csv("PS4data.csv") -> PS4data
cat("The dimensions of the dataset are:",
    dim(PS4data)[1], "rows and", dim(PS4data)[2], "columns.\n")
```

The dimensions of the dataset are: 3034 rows and 8 columns.

```
PS4data %>%
  filter(Country == "Turkey") %>%
  ggplot(aes(x = Year, y = Satis)) +
  geom_line(color= "lightpink", size=1 ) +
  geom_point(color= "deeppink2", size= 2) +
  labs(title = "Turkey's Democratic Satisfaction Over Time",
       x = "Year",
       y = "Democratic Satisfaction") +
  theme_minimal()+
  theme(
    plot.title = element_text(color = "deeppink3", face = "bold",
                              size = 12, hjust = 0.5),
    axis.title.x = element_text(color = "hotpink3",
                                face = "bold", size = 11),
    axis.title.y = element_text(color = "maroon2",
                                face = "bold", size = 11),
    panel.grid.major = element_line(color = "lavenderblush",alpha=0.005),
    panel.grid.minor = element_line(color = "lavenderblush",alpha=0.005),
    axis.line = ggplot2::element_line(color = "violetred"),
    axis.text = ggplot2::element_text(color = "palevioletred")
  )
```



What we can observe from the above plot is that there are significant yearly fluctuations in Turkey's democratic satisfaction. We initially observe a period of upward trend leading up to 2005, and from 2005 till 2015, we witness multiple fluctuations. During this decade, 2015 witnesses the lowest satisfaction score. However, after 2015, we see a positive trend that maintains stability and makes up for the previous year's variations.

Question 2: Then, estimate the following linear regression model where `DemSatis_it` is the democracy satisfaction score for country *i* in year *t*, `LiberalComponent_it` is V-Dem's `v2x_liberal` variable, and `ElectoralDemocracy_it` is the `v2x_polyarchy` variable for country *i* in year *t*. Show your code and print your results. What is your conclusion? Carefully interpret each estimated coefficient and tell whether they are significantly different from zero

```
vdem_sub <- vdem %>%
  select(
    country_name,
    year,
    v2x_liberal,
    v2x_polyarchy
  ) %>%
  mutate(country_name = ifelse(country_name == "Türkiye", "Turkey",
                                country_name))
```

```
#merging with ps4
merged_data <- PS4data %>%
  left_join(
    vdem_sub,
    by = c("Country" = "country_name", "Year" = "year"))

model_q2 <- lm(
  Satis ~ v2x_liberal + v2x_polyarchy,
  data = merged_data)

summary(model_q2)
```

Call:

```
lm(formula = Satis ~ v2x_liberal + v2x_polyarchy, data = merged_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.50502	-0.58950	-0.03062	0.60561	2.80287

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.64743	0.05859	-11.050	<2e-16 ***
v2x_liberal	1.16601	0.17062	6.834	1e-11 ***
v2x_polyarchy	-0.28935	0.16954	-1.707	0.088 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9327 on 2947 degrees of freedom

(84 observations deleted due to missingness)

Multiple R-squared: 0.0466, Adjusted R-squared: 0.04595

F-statistic: 72.02 on 2 and 2947 DF, p-value: < 2.2e-16

The coefficient for the intercept is -0.647, this value is supposed to represent our baseline for when the other regressors = 0. Thus, -0.647 is the predicted level of democratic satisfaction when electoral and liberal democracy components are equal to 0. This value is statistically significant with a p-value < 2e-16 under 5% significance level.

The coefficient for liberal democracy is 1.166. In other words, a 1 unit increase in the liberal democracy component is associated with a 1.16 unit increase in democratic satisfaction. In our context, it means that countries with high levels of liberal democracy (protection against

tyranny of the majority, importance of minority rights etc) report higher levels of democratic satisfaction. This value is also statistically significant with a p-value = $1e-11 < 0.001$. The coefficient for electoral democracy is -0.289. In other words, a 1 unit increase in electoral democracy actually causes 0.29 unit decrease in democratic satisfaction. However, this value is not significant as the p-value of $0.08 > 0.05$.

Question 3: Run a country-fixed effect regression by adding the Country variable into the model. Another, and more efficient, way to include country-fixed effects is to actually do the de-meaning process described above. Luckily, the fixest package does it for us. Use the feols function and estimate the parameters of the model specified above. That is, estimate the fixed-effect regression using this second approach with the feols function (you will have to figure out how to include country fixed effects). Then use the modelsummary package to summarize your results in a nice table. What is your conclusion based on this new model? Please interpret your estimated coefficients and tell whether they are statistically significant from zero. Show your code.

```
model_q3 <- feols(
  Satis ~ v2x_liberal + v2x_polyarchy | Country,
  data = merged_data
)
modelsummary(
  model_q3,
  coef_map = c(
    "v2x_polyarchy" = "Electoral democracy index",
    "v2x_liberal" = "Liberal component index"
  ),
  stars = TRUE,
  gof_map = c("nobs", "r.squared", "adj.r.squared"),
  notes = c(
    "Model includes country fixed effects.",
    "Fixed effects estimated using the within (de-meaning) estimator via feols.",
    "+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001"
  )
)
```

The coefficient for the liberal democracy component is now -0.881. Therefore, a 1 unit increase in liberal democracy actually results in a 0.8 unit decrease in the democratic satisfaction score. This value is also statistically significant with a p-value < 0.0001 . This new coefficient differs from our original model. Originally our coefficient was positive and now it is negative which shows us that this variable was impacted by a possible omitted variable bias and by holding country fixed effects, we see that liberal democracy actually has a negative impact on democratic satisfaction. The coefficient for electoral democracy is 0.379 which in our context

	(1)
Electoral democracy index	0.379+
	(0.212)
Liberal component index	−0.881***
	(0.204)
Num.Obs.	2950
R2	0.830
R2 Adj.	0.822
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

means a 1 unit increase is associated with a 0.37 unit increase in democratic score. While this value also differs from the coefficient in the original model, it is not highly significant. It's p-value is less than 0.1 so it only remains statistically significant at the 10% significance level.

Question 4: Now, estimate the same fixed effect model by adding the country variable as a factor instead of de-meaning. This would add the country variable as a set of dummy variables. Verify that you get the identical estimated coefficients as above. Show your code

```
model_q4 <- lm(
  Satis ~ v2x_liberal + v2x_polyarchy + factor(Country),
  data = merged_data
)

modelsummary(
  list(
    "Country FE (de-meaning, feols)" = model_q3,
    "Country FE (country dummies, lm)" = model_q4
  ),
  coef_map = c(
    "v2x_polyarchy" = "Electoral democracy index",
    "v2x_liberal" = "Liberal component index"
  ),
  stars = TRUE,
  gof_map = c("nobs", "r.squared", "adj.r.squared"))
```

	Country FE (de-meaning, feols)	Country FE (country dummies, lm)
Electoral democracy index	0.379+ (0.212)	0.379+ (0.212)
Liberal component index	-0.881*** (0.204)	-0.881*** (0.204)
Num.Obs.	2950	2950
R2	0.830	0.830
R2 Adj.	0.822	0.822

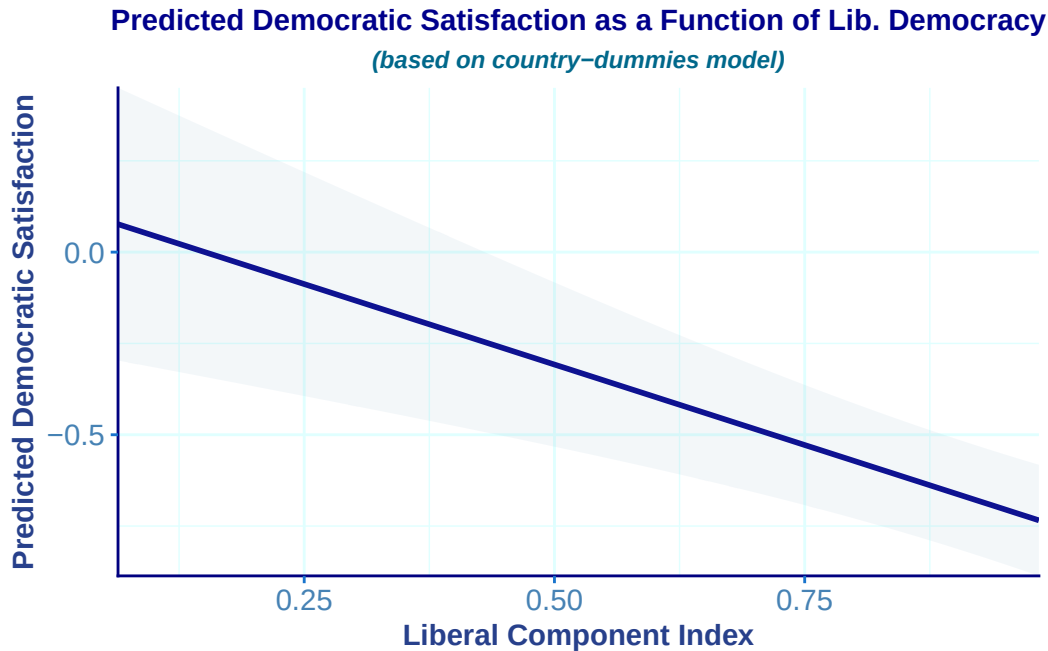
+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

As we can see from the table above, both models yield identical estimated coefficients for the liberal and electoral democracy indices. This confirms that both approaches effectively control for country fixed effects, leading to consistent results regardless of the method used.

Question 5: Carefully interpret the adjusted R^2 s you find in Question 2 (no fixed effects model) and 3 (fixed effects model). Why do you think that R^2 increased dramatically in the latter?

In Question 2, our R^2 value was 0.046, which means that our initial model only explained 4.6% of the variation. R^2 as a coefficient of determination, this value is quite low as it suggests that the remaining 95.4% is still unaccounted for. In Question 3, our R^2 value increases to 0.830, now explaining 83% of our total variation, but now our total variation was only within-country variation. This 78% difference can be attributed to the country fixed effects. In our model from Question 2, our model was not only trying to explain the difference in democratic scores within Turkey over the years but also cross-country differences, but after holding country fixed-effects, we removed all time-invariant cross country differences and only focused on within country differences. So our new model was then only responsible for explaining within country differences, leading to a higher R^2 value.

Question 6: Using the model you estimated in Question 4 (the one where you estimated the FE model with country dummies) and holding all other variables at their means, predict democracy satisfaction scores as a function of the liberal component index. Then, plot these predicted scores with their 95% confidence intervals and the liberal component index on the x-axis. *(PS doesnt say to show code)



The plot above illustrates the predicted democratic satisfaction scores as a function of the liberal component index, based on the country-dummies model. What we can see is steep downward trend, indicating that as the liberal component index increases, the predicted democratic satisfaction score decreases significantly. This suggests that higher levels of liberal democracy are associated with lower democratic satisfaction in this context. The shaded area represents the 95% confidence intervals, indicating the uncertainty around the predictions. We can see that the intervals go from relatively wide to narrow as we go from extreme liberal democracy values to moderate values.

Question 7: Extract Turkey's most recent liberal component index score. Based on the simulation in the previous section, how would the democracy satisfaction score change if Turkey's liberal component index increases to Sweden's liberal component index in 2020? Create a figure in which you put Turkey and hypothetical Turkey (the one with Sweden's liberal component index score) on the x-axis and predicted satisfaction score on the y axis. Make sure to have 95% confidence intervals around predictions. Show your code.

```
#turkey's most recent liberal component index score:
turkey_latest <- merged_data %>%
  dplyr::filter(Country == "Turkey") %>%
  dplyr::arrange(desc(Year)) %>%
  dplyr::slice(1) %>%
  dplyr::pull(v2x_liberal)
```

```

#swedens liberal component index in 2020
sweden_2020 <- merged_data %>%
  dplyr::filter(Country == "Sweden", Year == 2020) %>%
  dplyr::select(v2x_liberal) %>%
  dplyr::pull()

hypothetical_data <- data.frame(
  v2x_liberal = c(turkey_latest, sweden_2020),
  v2x_polyarchy = mean(merged_data$v2x_polyarchy, na.rm = TRUE),
  Country = "Turkey")

pred_hypothetical <- predict(
  model_q4,
  newdata = hypothetical_data,
  interval = "confidence",
  level = 0.95)
hypothetical_df <- cbind(hypothetical_data, pred_hypothetical)
hypothetical_df$Label <- factor(
  c("Turkey", "Hypothetical Turkey (Sweden's Liberal Index)"),
  levels = c("Turkey", "Hypothetical Turkey (Sweden's Liberal Index)"))

ggplot2::ggplot(hypothetical_df, ggplot2::aes(x = Label, y = fit)) +
  ggplot2::geom_point(size = 3, color = "blueviolet") +
  ggplot2::geom_errorbar(
    ggplot2::aes(ymin = lwr, ymax = upr),
    width = 0.2,
    color = "darkorchid",
    linewidth = 0.7
  ) +
  ggplot2::labs(
    title = "Predicted Democratic Satisfaction: Turkey vs. Hypothetical Turkey",
    subtitle = "Effect of Adopting Sweden's 2020 Liberal Component Index",
    x = "",
    y = "Predicted Democratic Satisfaction Score"
  ) +
  ggplot2::theme_classic(base_size = 11, base_family = "sans") +
  ggplot2::theme(
    plot.title = ggplot2::element_text(
      size = 11,
      face = "bold",
      hjust = 0.5,
      color = "purple4"
    )
  )

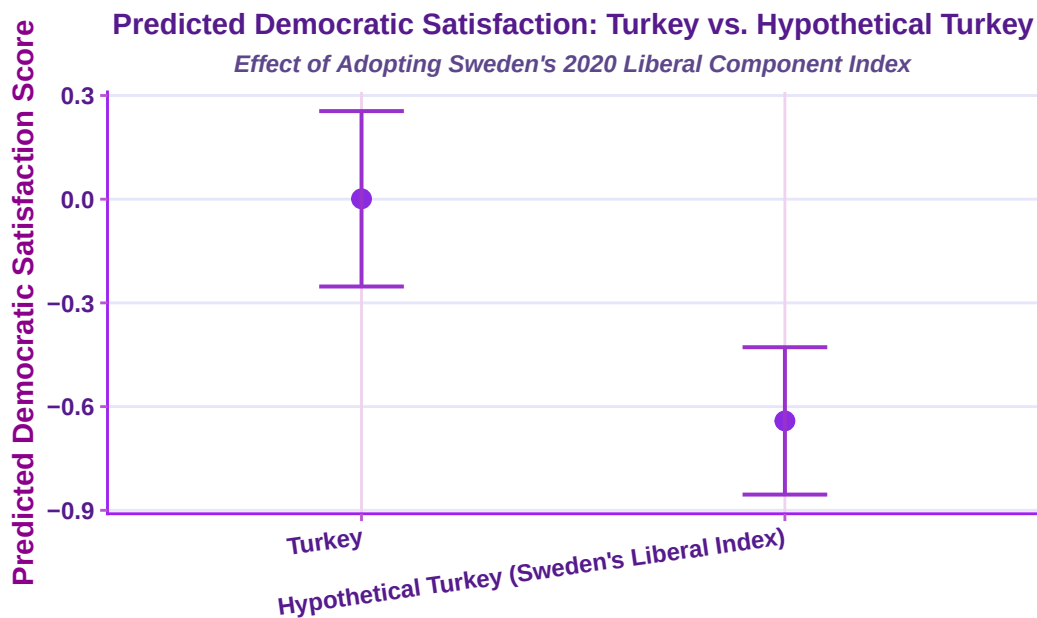
```



```

),
plot.subtitle = ggplot2::element_text(
  size = 9,
  hjust = 0.5,
  face = "bold.italic",
  color = "mediumpurple4"
),
axis.title.x = ggplot2::element_text(size = 11,
                                     color = "darkmagenta", face = "bold"),
axis.title.y = ggplot2::element_text(size = 11,
                                     color = "darkmagenta", face = "bold"),
axis.text = ggplot2::element_text(size = 9, face = "bold"),
axis.text.x = ggplot2::element_text(angle = 8, hjust = 1)
)+ ggplot2::theme(
  panel.grid.major.y = ggplot2::element_line(color = "lavender", alpha = 0.1),
  panel.grid.major.x = ggplot2::element_line(color = "thistle2", alpha = 0.1),
  axis.line = ggplot2::element_line(color = "purple"),
  axis.ticks = ggplot2::element_line(color = "mediumorchid"),
  axis.text = ggplot2::element_text(color = "purple4")
)

```



The plot above illustrates the predicted democratic satisfaction scores for Turkey based on its

current and hypothetical liberal component index. The first point represents Turkey's actual score, while the second point shows the predicted score if Turkey were to adopt Sweden's Liberal Component Index from 2020. The error bars indicate the 95% confidence intervals around these predictions, highlighting the uncertainty associated with each estimate. The comparison suggests that adopting Sweden's index could lead to a significant change in Turkey's democratic satisfaction score. What we can observe from the plot above is that if Turkey were to adopt Sweden's liberal component index, its predicted democratic satisfaction score would decrease significantly, by almost -0.6 units. This reaffirms our earlier finding from the model in Question 3 and 4, that higher levels of liberal democracy are associated with lower democratic satisfaction in Turkey's context.

Question 8: Although the fixed effect model controls for time-invariant heterogeneity, the exogeneity assumption can still fail if we omit time-variant variables that may correlate electoral democracy index and liberal component index. One possible confounder is GDP growth since it is well-known that economic crises cause regime changes. Therefore, we update the previous model by adding two macroeconomic variables: GDP growth (NY.GDP.MKTP.KD.ZG) and inflation (FP.CPI.TOTL.ZG). Using the WDI library, download these variables and merge them into the data you used above. Then, update the country-fixed effects model above by adding these two variables.

```
wdi_variables <- c(
  gdp_growth = "NY.GDP.MKTP.KD.ZG",
  inflation   = "FP.CPI.TOTL.ZG")

wdi_data <- WDI(
  country = "all",
  indicator = wdi_variables,
  start = min(merged_data$Year, na.rm = TRUE),
  end = max(merged_data$Year, na.rm = TRUE),
)
wdi_data <- wdi_data %>%
  rename(
    Country = country,
    Year     = year
  ) %>%
  select(Country, Year, gdp_growth, inflation)

merged_data_wdi <- merged_data %>%
  left_join(wdi_data, by = c("Country", "Year"))

model_q8 <- feols(
  Satis ~ v2x_liberal + v2x_polyarchy + gdp_growth + inflation | Country,
```

	(1)
Electoral democracy index	0.094 (0.229)
Liberal component index	−0.691** (0.234)
GDP growth	0.008*** (0.002)
Inflation	0.000 (0.000)
Num.Obs.	2594
R2	0.835
R2 Adj.	0.827
Country Fixed Effects	Yes
+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001	

```

data = merged_data_wdi
)
modelsummary(
  model_q8,
  stars = TRUE,
  coef_map = c(
    "v2x_polyarchy" = "Electoral democracy index",
    "v2x_liberal"    = "Liberal component index",
    "gdp_growth"     = "GDP growth",
    "inflation"      = "Inflation"
  ),
  gof_map = c("nobs", "r.squared", "adj.r.squared"),
  add_rows = data.frame(
    term = "Country Fixed Effects",
    estimate = "Yes"
  )
)

```

In this updated model, we have included GDP growth and inflation as additional control variables to account for potential time-variant confounders that may influence democratic satisfaction.

By doing this, we see that our coefficient for Liberal Democracy is -0.691, indicating that a 1 unit increase in Liberal Democracy would result in a 0.69 unit decrease in democratic satisfaction. This value is statistically significant with a p-value < 0.0001 . The coefficient for Electoral Democracy is 0.094, suggesting that a 1 unit increase in Electoral Democracy is associated with a 0.09 unit increase in democratic satisfaction. However, this value is not statistically significant as its p-value > 0.1 . Thus, we cannot confidently assert that Electoral Democracy has a meaningful impact on democratic satisfaction in this context. The coefficient for GDP growth is 0.008, indicating that a 1 percentage point increase in GDP growth is associated with a 0.008 unit increase in democratic satisfaction. This value is very highly significant with a p-value < 0.0001 , showing us that economic growth matters significantly for democratic satisfaction. The coefficient for inflation is -0.000, suggesting that inflation has no effect on democratic satisfaction. This value is not statistically significant as its p-value > 0.1 . Compared to our previous model without these controls, we see that the coefficient for liberal democracy decreased in magnitude - going from causing a 0.8 unit decrease in democratic satisfaction to a 0.69 unit decrease. This suggests that part of the relationship between liberal democracy and democratic satisfaction was confounded by economic factors. By controlling for GDP growth and inflation, we obtain a more accurate estimate of the true effect of liberal democracy on democratic satisfaction. Similarly, the coefficient for Electoral Democracy also decreased in magnitude, indicating that economic factors also confounded this relationship - going from causing a 0.3 unit increase in democratic satisfaction to a 0.09 unit increase, however, in both cases, these effects were not statistically significant. Overall, including GDP growth and inflation in our model helps to clarify the relationships and provides more reliable estimates of the effects of democratic components on satisfaction. Lastly, our adjusted R^2 increased slightly from 0.830 to 0.835, indicating that adding these macroeconomic controls improved the model's explanatory power but very slightly. This suggests that while economic factors do play a role in democratic satisfaction, the majority of the variation is still explained by the democratic components and country fixed effects.

Question 9: First, create an indicator variable that takes the value one if the country is a post-Soviet country and 0 otherwise. Then, use a two-sample t-test to test whether the average democracy satisfaction between post-Soviet countries and others is significantly different from each other in 1992 and 2020 (so two t-tests). Carefully interpret the results from these two t-tests. What is your overall conclusion? Do we reject the null hypothesis that post-Soviet citizens are equally satisfied with democracy as the rest of the world?

```
post_soviet_countries <- c(
  "Russia", "Ukraine", "Belarus", "Moldova",
  "Estonia", "Latvia", "Lithuania",
  "Georgia", "Armenia", "Azerbaijan",
  "Kazakhstan", "Uzbekistan", "Turkmenistan",
  "Kyrgyzstan", "Tajikistan"
```

```

)
merged_data <- merged_data %>%
  mutate(
    post_soviet = ifelse(Country %in% post_soviet_countries, 1, 0)
  )
data_1992 <- merged_data %>%
  filter(Year == 1992)

data_2020 <- merged_data %>%
  filter(Year == 2020)

t_test_1992 <- t.test(
  Satis ~ post_soviet,
  data = data_1992
)
ttxt <- capture.output(t_test_1992)
ttxt <- gsub(
  paste0(
    "alternative hypothesis: true difference in means between group 0 and ",
    "group 1 is not equal to 0"
  ),
  "alternative hypothesis: diff in means != 0",
  ttxt
)
cat(paste(ttxt, collapse = "\n"))

```

Welch Two Sample t-test

```

data: Satis by post_soviet
t = 4.0918, df = 22.725, p-value = 0.0004567
alternative hypothesis: diff in means != 0
95 percent confidence interval:
 0.6091746 1.8565951
sample estimates:
mean in group 0 mean in group 1
 -0.05723035    -1.29011517

```

```

t_test_2020 <- t.test(
  Satis ~ post_soviet,
  data = data_2020
)

```

```

)

ttxt1 <- capture.output(t_test_2020)
ttxt1 <- gsub(
  paste0(
    "alternative hypothesis: true difference in means between group 0 and ",
    "group 1 is not equal to 0"
  ),
  "alternative hypothesis: diff in means != 0",
  ttxt1
)
cat(paste(ttxt1, collapse = "\n"))

```

Welch Two Sample t-test

```

data: Satis by post_soviet
t = 2.0347, df = 14.139, p-value = 0.06108
alternative hypothesis: diff in means != 0
95 percent confidence interval:
 -0.03331471  1.28683331
sample estimates:
mean in group 0 mean in group 1
    0.08420146    -0.54255784

```

What we can observe from our two t-tests is that firstly, for 1992, the average democratic satisfaction score for post-Soviet countries is -1.29, while for non-post-Soviet countries, it is -0.05. The difference in means is 1.23, indicating that post-Soviet countries faced significantly lower democratic satisfaction compared to other countries in 1992. The p-value for this test is 0.0004 (<0.001) making it highly statistically significant and so we reject H_0 that post-soviet nations are equally satisfied with democracy as the rest of the world. For 2020, the average democratic satisfaction score for post-Soviet countries is -0.54, while for non-post-Soviet countries, it is 0.08. The difference in means is 0.627, indicating that even in 2020, post-Soviet countries still experience significantly lower democratic satisfaction compared to other countries. The p-value for this test is 0.061, so at the 5% significance level, we fail to reject H_0 , but at the 10% significance level, we can reject H_0 . This conclusion is reaffirmed by the 95% confidence interval that includes 0. In conclusion, even in 2020, there is evidence that post-Soviet are less satisfied with democracy, but the evidence is weak compared to 1992.